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Soap versus sanitiser for preventing the transmission of acute respiratory infections: a systematic review with meta-analysis and dose-response analysis

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Title page

Title

Soap versus sanitiser for preventing the transmission of acute respiratory infections: a systematic review with meta-analysis and dose-response analysis

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Abstract

Objective:

To compare the effectiveness of hand hygiene using alcohol-based hand sanitiser to soap and water for preventing the transmission of acute respiratory infections (ARIs), and assess the relationship between the dose of hand hygiene and the number of ARI, influenza-like illness (ILI), or influenza events.

Design:

Systematic review and meta-analysis

Data sources:

CENTRAL, PubMed, Embase, CINAHL and trial registries were searched in April 2020.

Inclusion criteria:

We included randomised controlled trials that compared a community-based hand hygiene intervention (soap and water, or sanitiser) with a control, or trials that compared sanitiser with soap and water, and measured outcomes of ARI, ILI, or laboratory-confirmed influenza or related consequences.

Data extraction and analysis:

Two review authors (MB, NK) independently screened the titles and abstracts for inclusion and extracted data.

Results:

Eighteen trials were included. When meta-analysed, three trials of soap and water versus control found a non-significant increase in ARI events (Risk Ratio (RR) 1.23, 95%CI 0.78-1.93); six trials of sanitiser versus control found a significant reduction in ARI events (RR 0.80, 95%CI 0.71-0.89). When hand hygiene dose was plotted against ARI relative risk, no clear dose-response relationship was observable. Four trials were head-to-head comparisons of sanitiser and soap and water but too heterogeneous to pool: two found a significantly greater reduction in the sanitiser group compared to the soap group; two found no significant difference between the intervention arms.

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Conclusions:

Adequately performed hand hygiene, with either soap or sanitiser, reduces the risk of ARI virus transmission, however direct and indirect evidence suggest sanitiser might be more effective in practice.

Key words:

Acute respiratory infections, Hand hygiene, Systematic review, dose-response analysis, Virus transmission.

Strengths and Limitations of this study

- Systematic review with meta-analysis and dose-response analysis of randomised controlled trials.
- Secondary analysis of a previously published systematic review and meta-analysis of physical interventions to reduce the spread of respiratory viruses
- Conclusions are mostly from indirectness evidence, with direct evidence available from only four head-to-head trials.

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Introduction

Acute respiratory infections (ARI) cause a substantial annual health burden, and much more so in the current COVID-19 pandemic. Globally, approximately 4 million deaths per year are caused by ARIs, with the current pandemic leading to just over 1 million deaths in 2020 already .[1, 2] To minimise the potential threat of ARIs to public health, the implementation of effective preventive community-based measures is essential.[3, 4]

Hand hygiene has previously been shown to be an effective intervention which reduces the transmission of the viruses and bacteria that cause ARIs.[5] It is a low-cost intervention that is applicable in all countries and all settings. Unlike vaccines which are disease-specific, hand hygiene has the advantage of being applicable to multiple ARIs.[5] Despite the effectiveness and worldwide applicability of hand hygiene, important questions for policy and practice remain, such as the dose-response and relative effectiveness of different materials (alcohol-based hand sanitiser; soap and water). This systematic review aimed to address these questions.

Methods

We aimed to find, appraise and synthesise studies of hand-hygiene interventions on ARIs in the community. This review forms a secondary analysis of a previously published systematic review and meta-analysis of physical interventions to reduce the spread of respiratory viruses [5, 6]. For the current systematic review, a sub-set of articles from the parent review relating to hand-hygiene interventions were used and analysed.

Inclusion criteria and study source

Participants

We included studies of participants of any age, gender or condition. Trials in healthcare settings were excluded.

Interventions

We included studies of interventions which compared a hand hygiene intervention (i.e. hand washing with soap and water or hand sanitiser) with a control intervention, or which compared two hand hygiene interventions.

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Outcomes (primary, secondary)

We only included studies that reported a measure of ARI – such as influenza-like illness, influenza, or respiratory infections – and this was our primary outcome. Studies were also eligible if they reported on ARI consequences (e.g. days off work, complications, hospitalisation or death, if clearly reported as consequences of the respiratory illness), as our secondary outcomes.

Study design

Randomised controlled trials (RCT) and cluster-randomised controlled trials (C-RCT) were eligible.

Search strategy

RCTs and C-RCTs studying the effectiveness of hand hygiene interventions in the transmission of ARIs were identified from the parent 2020 systematic review.[5] Studies for that review were identified via a search of the Cochrane Central Register of Controlled Trials, Embase and CINAHL, covering the dates October 2010 to 9 March 2020. The search string was designed for PubMed using the word frequency analyser and then translated for use in other databases using the Polyglot Search Translator.[7] A backwards and forward citation analysis, using Scopus, was conducted on all new studies retrieved. Search and citation analysis results were screened using the RobotSearch tool to remove all obvious non-RCTs.[8] While the analysis of the parent review was being conducted, a new Cochrane review of rinse-free handwashing in school and pre-school children was published and we also screened its included studies for possible eligible studies [9]. Three authors independently reviewed the titles and abstracts of identified studies to assess eligibility for inclusion. Discrepancies were resolved by consensus.

Screening and data extraction

Two review authors (MB, NK) independently screened the titles and abstracts for inclusion against the inclusion criteria. One author retrieved full-text, and three authors screened the full-texts for inclusion. Any disagreements were resolved by discussion, or reference to a third author. Data were independently extracted by two authors (MB, NK) on: volume or weight of material (e.g. sanitiser or soap) used per person per day, and number of

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handwashes per person per day. When not reported directly, we estimated usage where possible (see Table 1). For estimation purposes, we used data on the average amount of material used per person per handwash as reported; if data were not reported, we assumed 0.035 grams of soap or 1.5ml of liquid used per handwash [10]. A data extraction form for outcome data was piloted on two studies in the review. The following data were extracted from the parent systematic review [5]: 1) study characteristics; 2) risk of bias assessments; 3) type of handwashing intervention(s) (e.g. soap, sanitiser, gel); and 4) risk ratios (RR), log RR, and standard error values for ARI or ILI or influenza (including the outcome with most events from each study).

Risk of bias assessment

Risk of bias was assessed with the Cochrane Collaboration's Risk of Bias tool 1.[11] Author pairs from the parent review independently screened for: the method of random sequence generation and allocation concealment (selection bias), blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), outcome reporting (attrition bias), and selective reporting (reporting bias). Disagreements were resolved by discussion or a third assessor. For each item, risk was either 'high', 'low' or 'unclear'.

Data analysis

To assess the relationship between handwashes per person per day and the number of ARI or ILI or influenza events, we conducted the following analyses: 1) only studies whose number of handwashes could be estimated (regardless of the type of handwash material), subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap) and 2) all studies (whether or not the number of handwashes could be estimated), subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap). We used a Chi^2 test to test for subgroup interactions. Meta-analyses were conducted using *Review Manager 5*.

We used risk ratios for results, reporting the number of participants with an event. We undertook meta-analyses where data were sufficient to pool (when ≥2 studies or comparisons reported the same outcome). We used a random effects model. The individual was used as the unit of analysis, where possible. However, where data on the number of

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individuals with primary and secondary outcomes of interest were not available, we extracted and used data for the closest equivalent ratio, e.g. a rate ratio based on the ratio of total number of events (e.g. ARI events) in each group irrespective of the number per person. We contacted investigators to provide missing data where feasible. We used the I^2 statistic to measure heterogeneity among the included trials. Data were sufficient to conduct a subgroup analysis by comparison (soap vs sanitiser vs combination soap and sanitiser).

Results

The PRISMA flowchart (Figure 1) shows the number of trials identified from the 2020 parent systematic review [5] and other sources. Eighteen trials were assessed as eligible; four were head-to-head comparisons of hand sanitiser and soap and water [12-15] and 16 compared hand hygiene with a control [12, 14, 16-29]. Table 1 presents study characteristics.

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1st author (country, year)	Study design	Setting	Study period in months	Population (Age range)	Number of participants	Type of Hand Hygiene intervention*					Duration of intervention	Number of handwashes / day	Calculations used for the number of handwashes / day (if applicable)	Amount of soap or alcohol used (ml or g) / person / day	Calculations used for the amount of alcohol or soap in ml or g / person / day (if applicable)
						Staff training / education	Classroom activities	Sanitiser	Soap	Combined (Sanitiser & Soap)					
Azor-Martinez (Spain, 2018)	C-RCT	CCCs	7	Ch (0 to 3 y)	911 Ch; 24 CCC	✓	✓	✓	✓		8 mo	7 (sanitiser)	Each child used sanitiser between 6-8 times /day	10.5 ml	assume mean 7 uses per day, 1.5 ml mean per use
Biswas (Bangladesh, 2019)	C-RCT	Sch	4	Ch (5 to 10 y)	10,855 Ch	✓	✓	✓			10 w	2.87	We calculated the percentage of handwashing practices by dividing the number of handwashing instances by the total number of handwashing opportunities. There were 921 handwashing opportunities in intervention group (page 5). 604 (66%) school children in the intervention group washed their hands [at end line] & 5077 students in the intervention group. 4.3mls / person / day & assumed 1.5ml of sanitiser use per wash gives 2.87 washes / day	4.3 ml	N/A
Correa (Colombia, 2012)	C-RCT	CCC	8	Ch (0 to 5 y)	42 CCC	✓	✓				8 mo	5.261	We assumed that 1 alcohol-based hand-sanitiser push = 1 instance of handwash, and took into account the proportion of community and preschools, as in the calc for ml / person / day, gives 5.261 pushes (or handwashes) per day	3.16 ml	32 community centres and 10 preschools included the trial, distributed among the intervention & control arms; assumed the 32 vs 10 ratio to estimate the increase in the number of pushes: 10/42 * 4.5 for preschools + 32/42*5.5 for community centres = 5.261 average number of pushes overall. 5.261 pushes at 1000 ml / 1666 pushes = 3.16 ml
Cowling (Hong Kong, 2008)	C-RCT	HH	7	A and Ch (NR)	198 HH	✓			✓		9 d	NR		NR	
Cowling (Hong Kong, 2009)	C-RCT	HH	8	A and Ch (NR)	407 HH	✓			✓		1 w	NR		NR	
Hubner (Germany, 2010)	RCT	O	12	A (NR)	1230 A	✓	✓				1y	3.84	0.19 * 6 (assumed > 5 handwashes means 6); 0.598 * 4 (average of 3 to 5); 0.205 * 1.5 (average of 1 to 2) + 0.007 * 0 to get the mean number of handwashes per day across the whole group = 3.84	11.52 ml	0.19 * 6 (assume > 5 handwashes means 6); 0.598 * 4 (average of 3 to 5); 0.205 * 1.5 (average of 1 to 2) + 0.007 * 0 to get the mean number of handwashes per day across the whole group = 3.84 mean handwashes / person * 3 ml per handwash = 11.52
Larson (USA, 2010)	C-RCT	HH	20	A and Ch (NR)	509 HH	✓	✓				19 mo	1.77	44.2% in the Education group (not handwash group) reported using hand sanitiser "occasionally at some point during the study" and 56.9% of these reported using hand sanitiser 1-2x in previous 24 hours". 2.65ml / person / day at 1.5ml sanitiser per handwash = 1.77 handwashes / day	2.65 ml	4.5 people / household; 30 days / month; 358 mls handwash per month gives: 2.65 ml/person/day
Lennell (Sweden, 2008)[1,**]	C-RCT	CCC	7	Ch (mean age 3 y)	1517 Ch; 29 CCC	✓	✓	✓			30 w	4 (sanitiser)	N/A	6 ml	N/A

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Little (England, 2015)	RCT	Onl	26	A (≥18)	20,066 A	✓			✓	4 mo	7.48	0.04 * 1 avg + 0.10 * 3.5 avg + 0.22 * 5.5 avg + 0.26 * 8 avg + 0.38 * 10 (assuming >=10 at 10) = 7.48 avg	NR	
Nicholson (India, 2014)	C-RCT	HH	10	Ch (5 to 18 y)	NR	✓	✓		✓	41 w	16	The median soap consumption was found to be 45 g per household per week in control households compared with 235g in intervention households. 5.6 g / person / day at 0.35 g of soap per wash = 16 handwashes a day	5.60 g	235g/week; 6 people/household; gives: 5.595 / person / day
Pickering (Kenya, 2013)	C-RCT	Sch	2	Ch (2 to 13 y)	6 Sch	✓	✓	✓	✓	8 w	NR		NR	
Ram (Bangladesh, 2015)	RCT	HH	18	A (NR)	NR	✓			✓	10 d	5.89	2.06g soap / person / day at 0.35g of soap per wash = 5.89 washes / day	2.06 g	Fig 3 reports median / capita soap use in grams by day of enrollment, days 2 to 12. Median / capita soap use in grams on last day (day 12) was 2.06g
Roberts (Australia, 2000)	C-RCT	CCC	8	Ch (0 to 3 y)	23 CCC	✓	✓	✓		8 mo	NR		NR	
Sandora (USA, 2005)	C-RCT	HH	5	A (NR) and Ch (6 mo to 5 y)	292 families			✓		5 mo	5.2	N/A	5.2 ml	Assumed 1 ml/hand sanitiser use (may not be correct as it is one pump - but could have done >1pump of sanitiser each handwash), 5.2 uses/day/person
Savolainen-Kopra (Finland, 2012)[2,**]	C-RCT	O	18	A	683 A; 21 office work units; 6 corporations	✓		✓	✓	15-16 mo	6.1 (soap) & 6.9 (sanitiser)	N/A	NR	
Simmerman (Thailand, 2011)	C-RCT	HH	16	A and Ch (NR)	442 Ch; 1147 household members	✓			✓	3 w	4.7	N/A	7.71 ml	54 ml / person / week = 7.714 ml / person / day
Stebbins (USA, 2011)	C-RCT	Sch	14	Ch (NR)	3360 Ch; 10 Sch	✓	✓			1 flu season	2.4	N/A	1.44 ml	0.6 ml per use*2.4 times per day=1.44
Zomer (Netherlands, 2015)	C-RCT	CCC	7	Ch (6 m to 3.5 y)	545 Ch; 71 CCC	✓			✓	6 mo	NR		NR	

C-RCT: cluster randomized controlled trial, **RCT**: randomized controlled trial, **A:** Adults, **Ch:** Children, **CCC:** Childcare centres, **Sch:** Schools, **HH:** Households, **Onl:** Online, **O:** Offices, **N/A:** Not applicable, **NR:** Not reported

[1] This study was not included in the dose-response analysis as the infection was reported for any infection (not specific to respiratory infections) [2] Bathrooms at the workplaces were equipped with liquid hand soap (all arms)

*Only Hand Hygiene arms of the included studies, **Studies not included in the meta-analysis.

Table 1. Characteristics of included studies

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Risk of bias assessment

Most studies (95%) were unblinded due to the nature of the intervention, leading to a high risk of bias judgement (see Figures 2 and 3). Only one study was blinded to staff.[27] Blinding of outcome assessment and attrition bias was poor across 67% and 44% of studies respectively. Sequence generation and allocation concealment were had a low risk of bias assessment in about half of the studies. 50% of studies had unclear risk of bias for selective outcome reporting due to the lack of sufficient information.

Trials of hand sanitiser or soap and water versus control

Combining the three trials of soap and water hand hygiene versus control found a non-significant increase in ARI events: risk ratio 1.23 (95% CI 0.78 to 1.93) but with high heterogeneity (Figure 3). Combining the seven trials of hand sanitiser versus control found a significant reduction in ARI events: risk ratio 0.80 (95% CI 0.71 to 0.89), providing some indirect evidence in favour of hand sanitiser. Appendix A.1 shows the forest plot for all trials, regardless of whether number of handwashes could be estimated.

Dose-response relationship: hand hygiene frequency versus risk of respiratory infection (ARI, ILI or influenza)

Eleven of the trials provided sufficient information to estimate the dose of hand hygiene, which we converted to number of hand hygiene events per day. Plotted against the relative risk of ARIs, there is little dose-response relationship evident for hand sanitiser (Figure 4). The difference in effectiveness between hand sanitiser and soap and water does not appear to be explained by a difference in frequency. The cluster randomised trial by Little and colleagues [22] primarily used soap and water but also offered participants free hand sanitiser; only 18% report collecting the sanitiser.

Head-to-head trials of hand hygiene with hand sanitiser vs with soap and water

Four trials directly compared hand sanitiser with soap and water: two in childcare centres, one at a primary school, and one in workplaces. In a cluster randomised trial of children and staff in Swedish childcare centres, those at centres who were randomised to use an alcohol-based oily disinfectant gel (70% ethanol) after regular hand washing had a reduction in

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absenteeism rate of 12% (95% CI 4% to 20%) compared to control centres which used only soap and water [13] . The 3-arm cluster randomised trial of 24 childcare centres in Spain - educational and hand hygiene measures (one with soap and water; another with hand sanitiser) and a control group found children in the sanitiser group had a 13% lower (95% CI 6% to 28%) risk of respiratory infection than children in the soap and water group [12].

In Kenya, a cluster randomised trial assigned two primary schools to receive a handwashing with soap and water intervention, two to receive a sanitiser intervention, and two were a control [14]. Compared to control group students, both intervention groups had a reduction in observed rhinorrhea (RR 0.77, 95% CI 0.62-0.95 for both sanitiser vs control and soap vs control). No significant differences between the sanitiser and soap groups were observed for respiratory outcomes. The 3-arm trial in six companies in Finland randomised workplaces to equip workplace bathrooms with liquid hand soap (soap and control arms) or alcohol-based hand rub [15]. Participants in the intervention arms also received guidance on additional strategies for limiting infection transmission. Before the onset of the 2009 influenza pandemic (and the subsequent national hand hygiene campaign), a statistically significant ($p = 0.002$) difference in the infection episodes was observed between the control (6.0 per year) and the soap-and-water arm (5.0 per year), but not between the control and the alcohol-rub arm (5.6 per year). Neither intervention had an effect on work absenteeism.

Discussion

Based on both indirect and direct (head-to-head) trials, hand hygiene using alcohol-based hand sanitiser appears more effective at reducing ARI transmission than hand hygiene using with soap and water, with the difference in effect not explained by the difference in frequency of hand hygiene. The apparent greater effectiveness of hand sanitiser may be explained by its greater convenience, less time required to perform hand hygiene, more sustained compliance with hand hygiene, and less irritation to the skin [30].

Limitations of this review are that conclusions are mostly from indirectness evidence, with direct evidence available from only four head-to-head trials, and that it was not possible to estimate the dose of hand hygiene for some trials.

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A recent Cochrane review of the effect of rinse-free handwashing, compared to traditional hand hygiene, on absenteeism for ARI in preschool and school children reported a significant reduction in absenteeism of 9 days per 1000 available days for children in the rinse-free group, with the results coming from six randomised trials [9]. The effectiveness of handwashing with materials other than sanitiser or soap and water, such as ash, which may be used in low-income countries has mostly been examined in observational studies with uncertain effects [31].

Conclusions

Hand hygiene has a modest but important role in reducing the transmission of ARIs. Adequately performed hand hygiene, with either soap or sanitiser, reduces the risk of acute respiratory virus transmission. However, from both the direct and indirect comparisons in this review, sanitiser appears more effective in practice. While further head-to-head randomised trials are warranted, the current evidence appears sufficient to promote the use of hand sanitiser as the primary means for many everyday situations.

Abbreviations: Acute respiratory infections (ARI), Influenza-like illness (ILI), Risk Ratio (RR), Confidence interval (CI)

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Contributors:

TH, PG, and CDM conceived the study. MB and NK screened studies for inclusion in this review and extracted the data. TH, AMS, PG and CDM analysed the data. TH wrote the first draft. All authors approved and revised the final manuscript.

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Competing interests:

None declared.

Ethical approval:

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Data sharing

All data relevant to the study are included in the article or uploaded as supplementary information.

Transparency

The corresponding authors affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

Patient and public involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

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Figure Caption List

Figure 1. PRISMA Flow Chart

Figure 2. Overall risk of bias - presented as percentages across all included RCTs.

Figure 3. Meta-analysis of studies whose number of handwashes could be estimated, subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)

Figure 4. Hand hygiene frequency ('dose') versus risk of respiratory infection (ARI, ILI or influenza)

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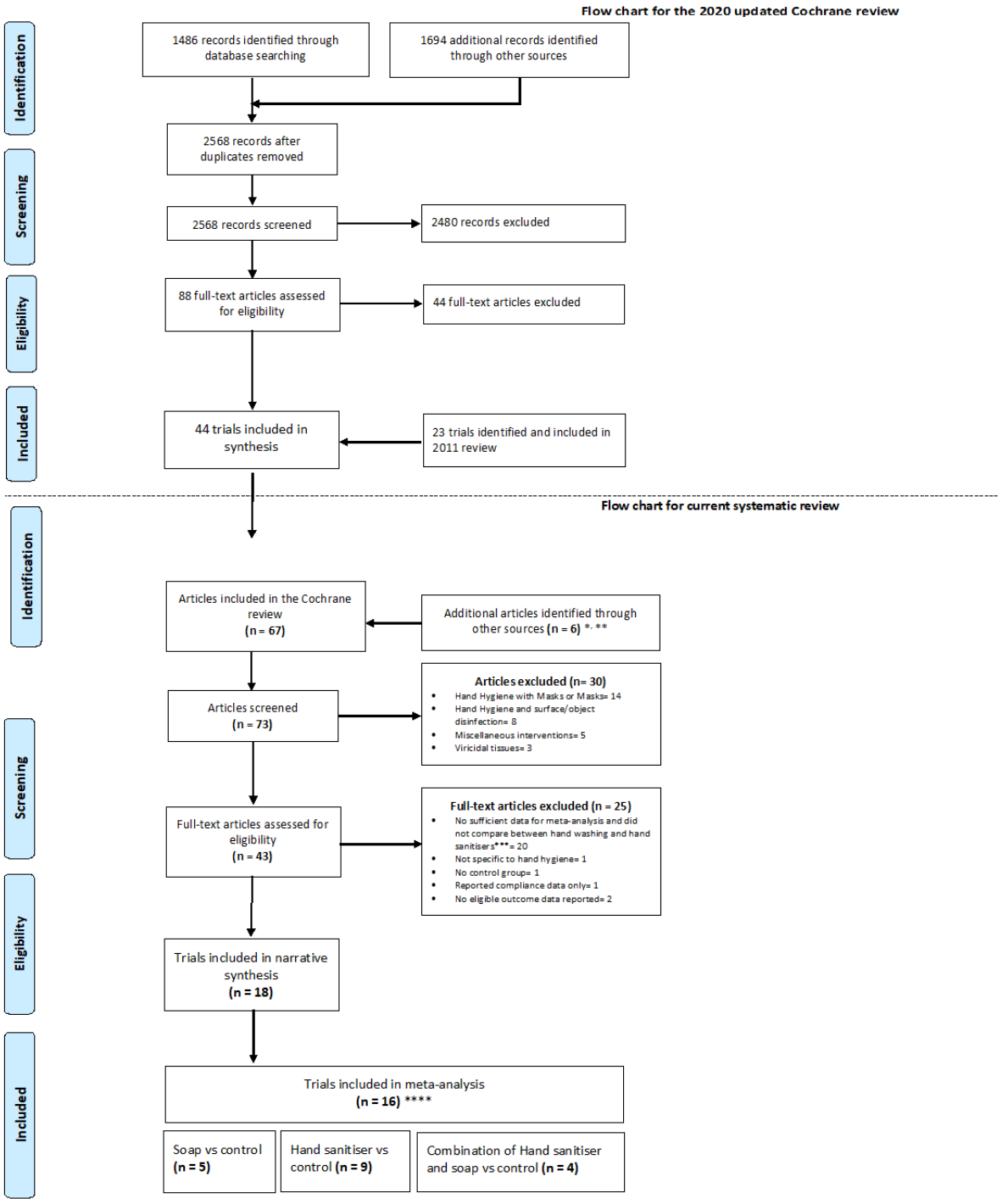
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Appendix A. Supplementary data

Appendix A1 – Forest plot of meta-analysis of all studies (regardless of whether the number of handwashes could be estimated) regardless of the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)

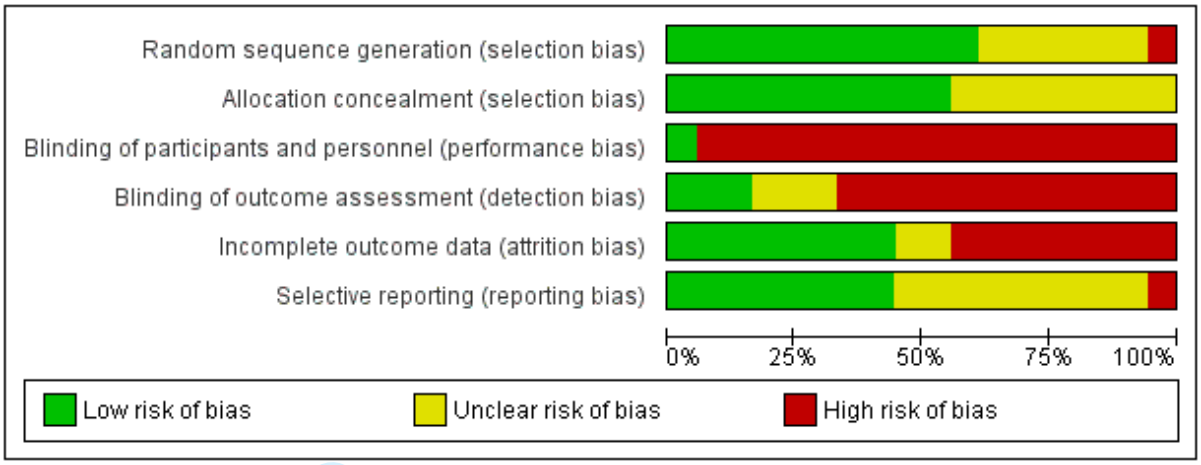
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Figure 1. PRISMA Flow Chart



* Articles identified by screening the references of a newly published systematic review (Munn et al. 2020).
** One of the identified additional studies (Lennell et al. 2008) was excluded from the Cochrane review, because no respiratory illness data could be extracted. However, it was included in this review as it reported head to head comparison between hand washing and hand sanitiser and it contained absenteeism data which was an eligible outcome for this review.
*** Two studies did not have sufficient data for meta-analysis, but were included in the head to head comparison between hand washing and hand sanitiser (Lennell et al. 2008 & Savolainen-Kopra et al. 2012)
**** Two studies (Azor-Martinez et al. 2018 & Pickering et al. 2013) had 2 intervention groups. 1 group used soap and water, and another used hand sanitiser. Therefore, the number of studies, split by type of hand hygiene intervention, adds up to 18 and not 16 studies.

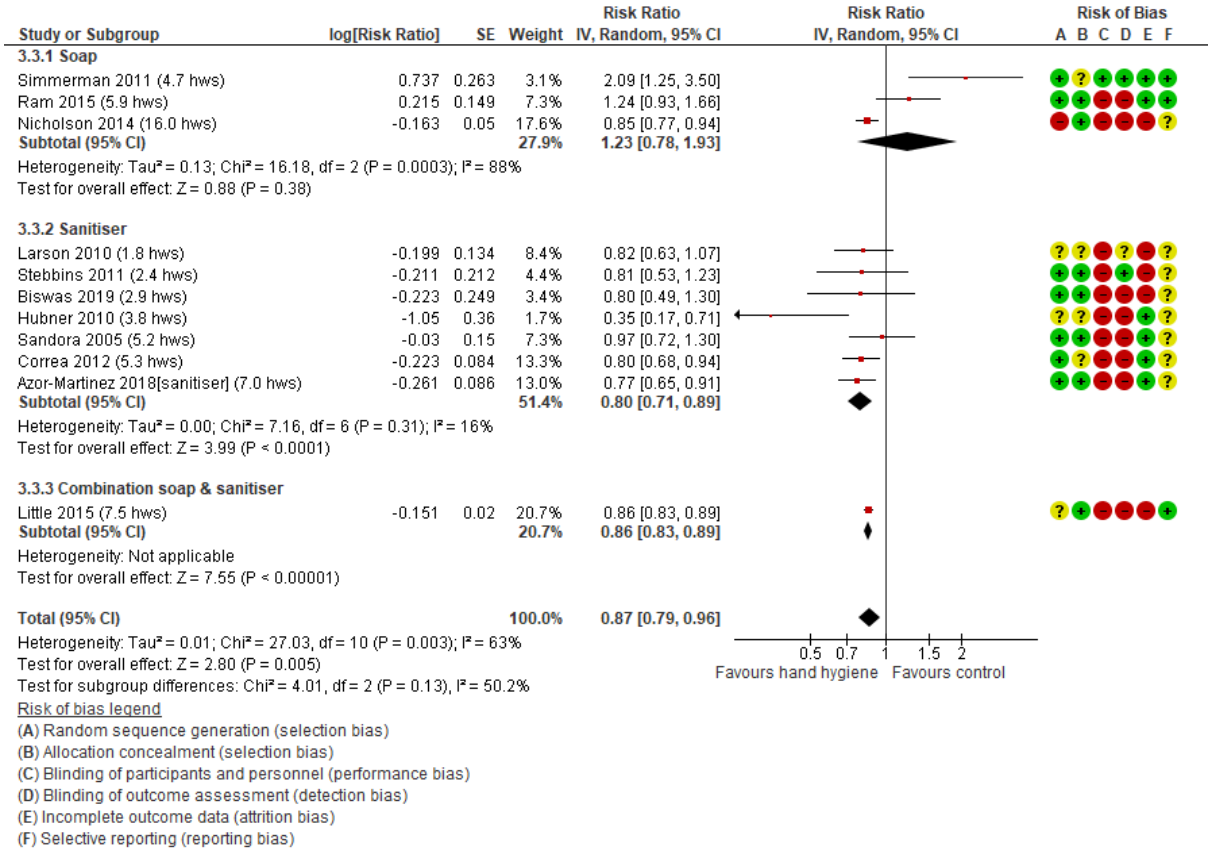
Figure 2. Overall risk of bias - presented as percentages across all included RCTs.



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Figure 3. Meta-analysis of studies whose number of handwashes could be estimated, subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)

[Coloured figure in print]

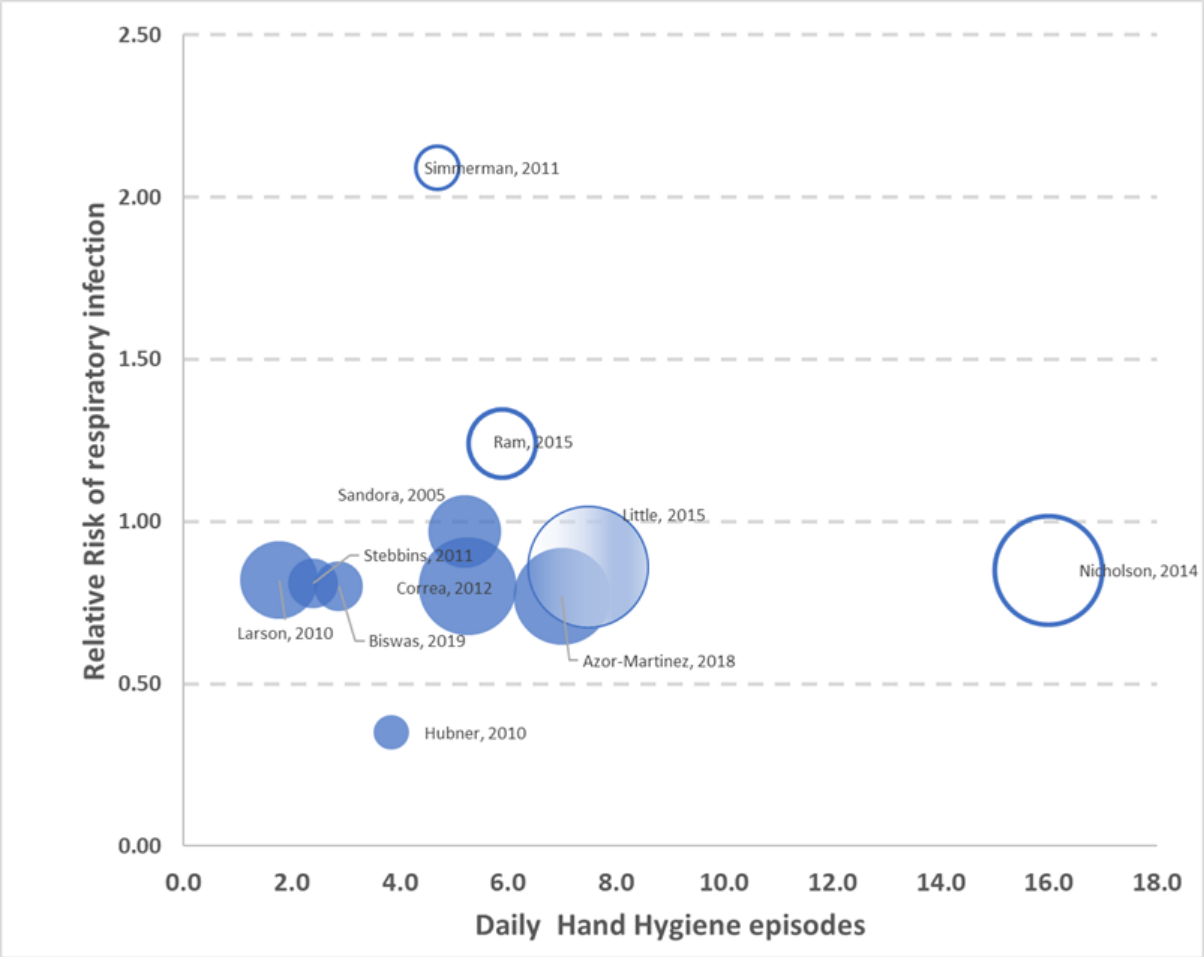


Number of handwashes (hws) indicated in brackets after study reference where estimable.

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Figure 4. Hand hygiene frequency ('dose') versus risk of respiratory infection (ARI, ILI or influenza)

[Coloured figure in print]



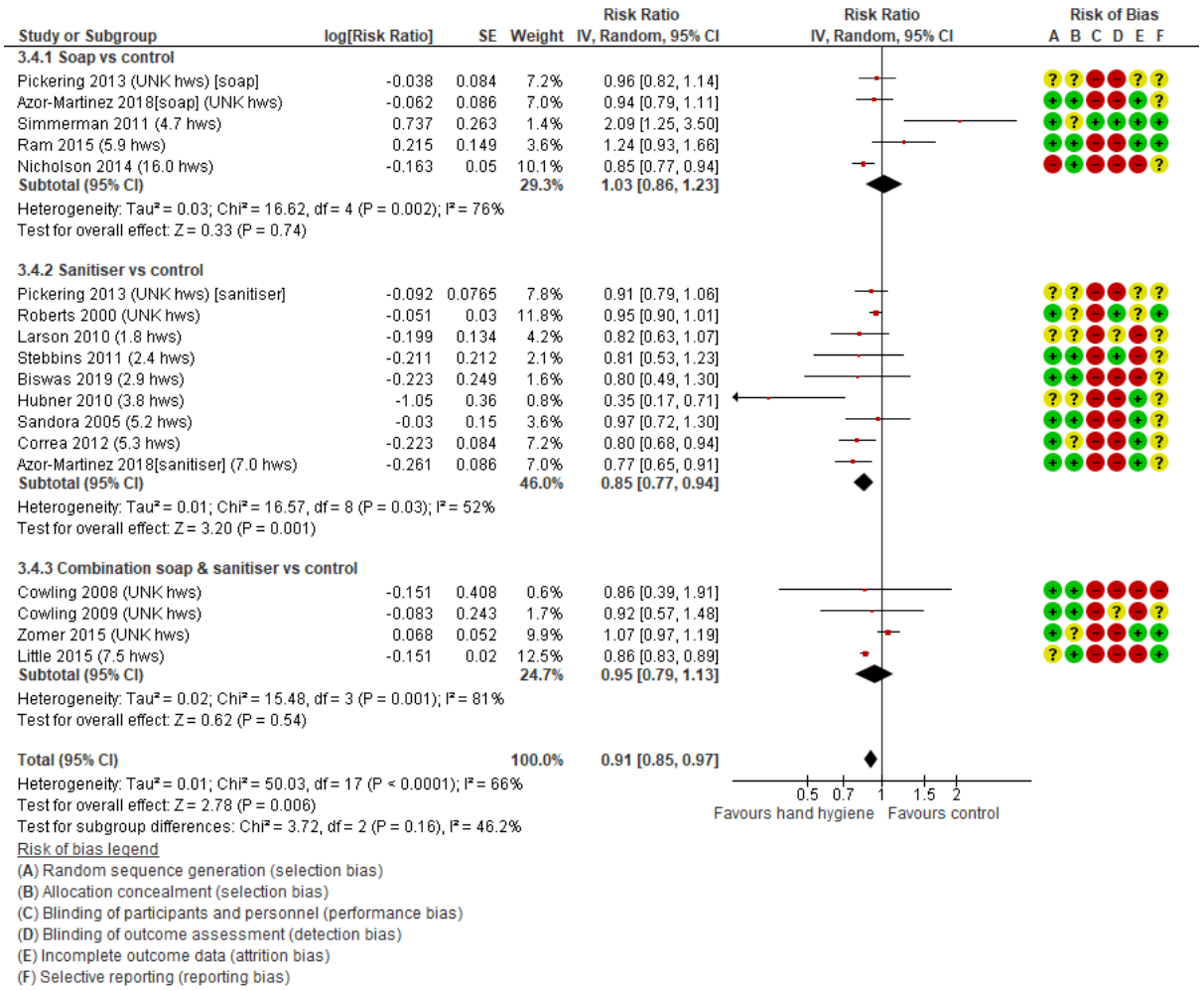
Clear circles = studies using soap

Coloured-in circles = studies using hand sanitiser

only

Appendix A

Appendix A.1 Forest plot of meta-analysis of all trials (regardless of whether the number of handwashes could be estimated), subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)



Number of handwashes (hws) indicated in brackets after study reference where estimable. UNK = the number of handwashes was not estimable.

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4, 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4, 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4, 5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5, 6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5, 6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4, 5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	-
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Fig 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Figures
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7, 8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Fig 2 & Appendix A1
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	-
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	10

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Soap versus sanitiser for preventing the transmission of acute respiratory infections in the community: a systematic review with meta-analysis and dose-response analysis

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Title page

Title

Soap versus sanitiser for preventing the transmission of acute respiratory infections in the community: a systematic review with meta-analysis and dose-response analysis

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Word count: 2478

Figures: 5

Tables: 1

Supplementary material: 3

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Abstract

Objective:

To compare the effectiveness of hand hygiene using alcohol-based hand sanitiser to soap and water for preventing the transmission of acute respiratory infections (ARIs), and to assess the relationship between the dose of hand hygiene and the number of ARI, influenza-like illness (ILI), or influenza events.

Design:

Systematic review and meta-analysis

Data sources:

CENTRAL, PubMed, Embase, CINAHL and trial registries were searched in April 2020.

Inclusion criteria:

We included randomised controlled trials that compared a community-based hand hygiene intervention (soap and water, or sanitiser) with a control, or trials that compared sanitiser with soap and water, and measured outcomes of ARI, ILI, or laboratory-confirmed influenza or related consequences.

Data extraction and analysis:

Two review authors independently screened the titles and abstracts for inclusion and extracted data.

Results:

Eighteen trials were included. When meta-analysed, three trials of soap and water versus control found a non-significant increase in ARI events (Risk Ratio (RR) 1.23, 95%CI 0.78-1.93); six trials of sanitiser versus control found a significant reduction in ARI events (RR 0.80, 95%CI 0.71-0.89). When hand hygiene dose was plotted against ARI relative risk, no clear dose-response relationship was observable. Four trials were head-to-head comparisons of sanitiser and soap and water but too heterogeneous to pool: two found a significantly greater reduction in the sanitiser group compared to the soap group; two found no significant difference between the intervention arms.

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Conclusions:

Adequately performed hand hygiene, with either soap or sanitiser, reduces the risk of ARI virus transmission, however direct and indirect evidence suggest sanitiser might be more effective in practice.

Key words:

Acute respiratory infections, Hand hygiene, Systematic review, dose-response analysis, Virus transmission.

Strengths and Limitations of this study

- Systematic review with meta-analysis and dose-response analysis of randomised controlled trials.
- Additional analysis of a subset of trials from a previously published systematic review and meta-analysis of physical interventions to reduce the spread of respiratory viruses
- Conclusions are mostly from indirectness evidence, with direct evidence available from only four head-to-head trials.

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Introduction

Acute respiratory infections (ARI) cause a substantial annual health burden, and much more so in the current COVID-19 pandemic. Globally, approximately 4 million deaths per year are caused by ARIs, with the current pandemic leading to just over 2.7 million deaths.[1, 2] To minimise the potential threat of ARIs to public health, the implementation of effective preventive community-based measures is essential.[3, 4]

Hand hygiene has previously been shown to be an effective intervention which reduces the transmission of the viruses and bacteria that cause ARIs.[5, 6] It is a low-cost intervention that is applicable in all countries and all settings. Unlike vaccines which are disease-specific, hand hygiene has the advantage of being applicable to multiple ARIs.[5] Despite the effectiveness and worldwide applicability of hand hygiene, important questions for policy and practice remain, such as the dose-response and relative effectiveness of different materials (alcohol-based hand sanitiser; soap and water). This systematic review aimed to address these questions. This review is an additional analysis of a subset of randomised trials that were included in a large systematic review that addressed all physical interventions (for example, face masks, personal protection, hand hygiene, and quarantining) to prevent the transmission of ARIs.[5]

Methods

We aimed to find, appraise, and synthesise studies of the effectiveness of hand hygiene interventions in the community for preventing ARI transmission. For this systematic review, a subset of articles relating to hand hygiene interventions were identified from a recently updated and published systematic review of all physical interventions for preventing ARI transmission. This review was not registered on PROSPERO as this review is an additional analysis of a subset of trials from a published Cochrane systematic review.[5]

Inclusion criteria and study source

Participants

We included studies of participants of any age, gender, or condition. Trials in healthcare settings were excluded.

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Interventions

We included studies of interventions which compared a hand hygiene intervention (i.e. hand washing with soap and water or hand sanitiser) with a control intervention, or which compared two hand hygiene interventions.

Outcomes

We only included studies that reported a measure of ARI – such as influenza-like illness, influenza, or respiratory infections – and this was our primary outcome. Studies were also eligible if they reported on ARI consequences (e.g. days off work, complications, hospitalisation or death, if clearly reported as consequences of the respiratory illness) and these were our secondary outcomes.

Study design

Randomised controlled trials (RCT) and cluster-randomised controlled trials (C-RCT) were eligible.

Search strategy

RCTs and C-RCTs studying the effectiveness of hand hygiene interventions in the transmission of ARIs were identified from the parent 2020 systematic review.[5] Studies for that review were identified via a search of the Cochrane Central Register of Controlled Trials, Embase and CINAHL, covering the dates October 2010 to 9 March 2020. The search string (see online supplemental File 1) was designed for PubMed using the word frequency analyser and then translated for use in other databases using the Polyglot Search Translator.[7] A backwards and forward citation analysis, using Scopus, was conducted on all new studies retrieved. Search and citation analysis results were screened using the RobotSearch tool to remove all obvious non-RCTs.[8] While the analysis of the parent review was being conducted, a new Cochrane review of rinse-free handwashing in school and pre-school children was published and we also screened its included studies for possible eligible studies [9]. Three authors (TH, MB, NK) independently reviewed the titles and abstracts of identified studies to assess eligibility for inclusion. Discrepancies were resolved by consensus.

Screening and data extraction

Two review authors (MB, NK) independently screened the titles and abstracts for inclusion

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against the inclusion criteria. One author retrieved the full text and two authors screened the full-texts for inclusion. Any disagreements were resolved by discussion between the authors or with a third author. Data were independently extracted by two authors (MB, NK) on: volume or weight of material (e.g. sanitiser or soap) used per person per day, and number of handwashes per person per day. When not reported directly, we estimated usage where possible (see Table 1). For estimation purposes, we used data on the average amount of material used per person per handwash as reported; if data were not reported, we assumed 0.035 grams of soap or 1.5ml of liquid used per handwash [10]. A data extraction form for outcome data was piloted on two studies in the review. The following data were extracted from the parent systematic review [5]: 1) study characteristics; 2) risk of bias assessments; 3) type of handwashing intervention(s) (e.g. soap, sanitiser, gel); and 4) risk ratios (RR), log RR, and standard error values for ARI or ILI or influenza (including the outcome with most events from each study).

Risk of bias assessment

Risk of bias was assessed with the Cochrane Collaboration's Risk of Bias tool 1.[11] Author pairs from the parent review independently screened for: the method of random sequence generation and allocation concealment (selection bias), blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), outcome reporting (attrition bias), and selective reporting (reporting bias). Disagreements were resolved by discussion or a third assessor. For each item, risk was either 'high', 'low' or 'unclear'.

Data analysis

To assess the relationship between handwashes per person per day and the number of ARI or ILI or influenza events, we conducted the following analyses: 1) only studies whose number of handwashes could be estimated (regardless of the type of handwash material), subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap) and 2) all studies (whether or not the number of handwashes could be estimated), subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap). We used a Chi^2 test to test for subgroup interactions. Meta-analyses were conducted using *Review Manager 5.4*.

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We used risk ratios for results, reporting the number of participants with an event. We undertook meta-analyses where data were sufficient to pool (when ≥2 studies or comparisons reported the same outcome). A random effects model was used because we expected some heterogeneity in the populations, interventions, and outcomes of the included studies. The individual was used as the unit of analysis, where possible. However, where data on the number of individuals with primary and secondary outcomes of interest were not available, we extracted and used data for the closest equivalent ratio, e.g. a rate ratio based on the ratio of total number of events (e.g. ARI events) in each group irrespective of the number per person. We contacted investigators to provide missing data where feasible. We used the I^2 statistic to measure heterogeneity among the included trials. We referred to the Cochrane Handbook in the interpretation of the I^2 statistic and were guided by the following ranges: 0% to 40%: may represent non-significant heterogeneity; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity.[11] Data were sufficient to conduct a subgroup analysis by comparison (soap vs sanitiser vs combination soap and sanitiser). To ensure we accounted for the clustering effect in the cluster-RCTs, we extracted the appropriate cluster-adjusted estimates from the trial reports and used the generic inverse variance method of meta-analysis.

Results

The PRISMA flowchart (Figure 1) shows the number of trials identified from the 2020 parent systematic review [5] and other sources. Eighteen trials were assessed as eligible; four were head-to-head comparisons of hand sanitiser and soap and water [12-15] and 16 compared hand hygiene with a control [12, 14, 16-29]. Table 1 presents study and intervention characteristics. Online supplemental File 2 contains a summary of the comparator conditions. The majority of trials used existing handwashing practices as the comparator condition. One trial additionally provided education on hand hygiene, four provided health promotion education and provided participants with education on the prevention and treatment of respiratory tract infections. None of the studies provided control patients with any hand hygiene materials.

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187 *[Insert Figure 1]*

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1st author (country, year)	Study design	Setting	Study period in months	Population (Age range)	Number of participants	Type of Hand Hygiene intervention*				Duration of intervention	Number of handwashes / day	Type of infection	Calculations used for the number of handwashes / day (if applicable)	Amount of soap or alcohol used (ml or g) / person / day	Calculations used for the amount of alcohol or soap in ml or g / person / day (if applicable)
						Staff training / education	Classroom activities	Sanitiser	Soap / Combined (Sanitiser & Soap)						
Azor-Martinez (Spain, 2018)	C-RCT	CCCs	7	Ch (0 to 3 y)	911 Ch; 24 CCC	✓	✓	✓	✓	8 mo	7 (sanitiser)	ARI	Each child used sanitiser between 6-8 times /day	10.5 ml	assume mean 7 uses per day, 1.5 ml mean per use
Biswas (Bangladesh, 2019)	C-RCT	Sch	4	Ch (5 to 10 y)	10,855 Ch	✓	✓	✓		10 w	2.87	Influenza	We calculated the percentage of handwashing practices by dividing the number of handwashing instances by the total number of handwashing opportunities. There were 921 handwashing opportunities in intervention group (page 5). 604 (66%) school children in the intervention group washed their hands [at end line] & 5077 students in the intervention group. 4.3mls / person / day & assumed 1.5ml of sanitiser use per wash gives 2.87 washes / day	4.3 ml	N/A
Correa (Colombia, 2012)	C-RCT	CCC	8	Ch (0 to 5 y)	42 CCC	✓	✓			8 mo	5.261	ARI & GI	We assumed that 1 alcohol-based hand-sanitiser push = 1 instance of handwash, and took into account the proportion of community and preschools, as in the calc for ml / person / day, gives 5.261 pushes (or handwashes) per day	3.16 ml	32 community centres and 10 preschools included the trial, distributed among the intervention & control arms; assumed the 32 vs 10 ratio to estimate the increase in the number of pushes: 10/42 * 4.5 for preschools + 32/42*5.5 for community centres = 5.261 average number of pushes overall. 5.261 pushes at 1000 ml / 1666 pushes = 3.16 ml
Cowling (Hong Kong, 2008)	C-RCT	HH	7	A and Ch (NR)	198 HH	✓			✓	9 d	NR	Influenza		NR	
Cowling (Hong Kong, 2009)	C-RCT	HH	8	A and Ch (NR)	407 HH	✓			✓	1 w	NR	Influenza		NR	
Hubner (Germany, 2010)	RCT	O	12	A (NR)	1230 A	✓	✓			1y	3.84	ARI & GI	0.19 * 6 (assumed > 5 handwashes means 6); 0.598 * 4 (average of 3 to 5); 0.205 * 1.5 (average of 1 to 2) + 0.007 * 0 to get the mean number of handwashes per day across the whole group = 3.84	11.52 ml	0.19 * 6 (assume > 5 handwashes means 6); 0.598 * 4 (average of 3 to 5); 0.205 * 1.5 (average of 1 to 2) + 0.007 * 0 to get the mean number of handwashes per day across the whole group = 3.84 mean handwashes / person * 3 ml per handwash = 11.52
Larson (USA, 2010)	C-RCT	HH	20	A and Ch (NR)	509 HH	✓	✓			19 mo	1.77	ILI & Influenza	44.2% in the Education group (not handwash group) reported using hand sanitiser "occasionally at some point during the study" and 56.9% of these reported using hand sanitiser 1-2x in previous 24 hours". 2.65ml / person / day at 1.5ml sanitiser per handwash = 1.77 handwashes / day	2.65 ml	4.5 people / household; 30 days / month; 358 mls handwash per month gives: 2.65 ml/person/day
Lennell (Sweden, 2008)[1, **]	C-RCT	CCC	7	Ch (mean age 3 y)	1517 Ch; 29 CCC	✓	✓	✓		30 w	4 (sanitiser)	NC	N/A	6 ml	N/A

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Little (England, 2015)	RCT	Onl	26	A (≥18)	20,066 A	✓			✓	4 mo	7.48	ARI	0.04 * 1 avg + 0.10 * 3.5 avg + 0.22 * 5.5 avg + 0.26 * 8 avg + 0.38 * 10 (assuming >=10 at 10) = 7.48 avg	NR	
Nicholson (India, 2014)	C-RCT	HH	10	Ch (5 to 18 y)	NR	✓	✓	✓		41 w	16	ARI & GI	The median soap consumption was found to be 45 g per household per week in control households compared with 235g in intervention households. 5.6 g / person / day at 0.35 g of soap per wash = 16 handwashes a day	5.60 g	235g/week; 6 people/household; gives: 5.595 / person / day
Pickering (Kenya, 2013)	C-RCT	Sch	2	Ch (2 to 13 y)	6 Sch	✓	✓	✓	✓	8 w	NR	ARI & GI		NR	
Ram (Bangladesh, 2015)	RCT	HH	18	A (NR)	NR	✓		✓		10 d	5.89	ILI & Influenza	2.06g soap / person / day at 0.35g of soap per wash = 5.89 washes / day	2.06 g	Fig 3 in Ram et al reports median / capita soap use in grams by day of enrollment, days 2 to 12. Median / capita soap use in grams on last day (day 12) was 2.06g
Roberts (Australia, 2000)	C-RCT	CCC	8	Ch (0 to 3 y)	23 CCC	✓	✓	✓		8 mo	NR	ARI		NR	
Sandora (USA, 2005)	C-RCT	HH	5	A (NR) and Ch (6 mo to 5 y)	292 families		✓			5 mo	5.2	ARI & GI	N/A	5.2 ml	Assumed 1 ml/hand sanitiser use (may not be correct as it is one pump - but could have done >1pump of sanitiser each handwash), 5.2 uses/day/person
Savolainen-Kopra (Finland, 2012)[2,**]	C-RCT	O	18	A	683 A; 21 office work units; 6 corporations	✓		✓	✓	15-16 mo	6.1 (soap) & 6.9 (sanitiser)	ARI & GI	N/A	NR	
Simmerman (Thailand, 2011)	C-RCT	HH	16	A and Ch (NR)	442 Ch; 1147 household members	✓		✓		3 w	4.7	Influenza	N/A	7.71 ml	54 ml / person / week = 7.714 ml / person / day
Stebbins (USA, 2011)	C-RCT	Sch	14	Ch (NR)	3360 Ch; 10 Sch	✓	✓			1 flu season	2.4	Influenza	N/A	1.44 ml	0.6 ml per use*2.4 times per day=1.44
Zomer (Netherlands, 2015)	C-RCT	CCC	7	Ch (6 m to 3.5 y)	545 Ch; 71 CCC	✓			✓	6 mo	NR	ARI & GI		NR	

C-RCT: cluster randomized controlled trial, **RCT**: randomized controlled trial, **A:** Adults, **Ch:** Children, **CCC:** Childcare centres, **Sch:** Schools, **HH:** Households, **Onl:** Online, **O:** Offices, **N/A:** Not applicable, **NR:** Not reported, **NC:** Not clear, **ARI:** Acute respiratory infections, **GI:** Gastrointestinal infections, **ILI:** Influenza-like illness

[1] This study was not included in the dose-response analysis as the infection was reported for any infection (not specific to respiratory infections) [2] Bathrooms at the workplaces were equipped with liquid hand soap (all arms)

*Only Hand Hygiene arms of the included studies, **Studies not included in the meta-analysis.

Table 1. Characteristics of included studies

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Risk of bias assessment

Most studies (95%) were unblinded due to the nature of the intervention, leading to a high risk of bias judgement (Figure 2). See online supplemental file 3 for risk of bias assessment for each individual trial. Only one study was blinded to staff.[27] Blinding of outcome assessment and attrition bias was poor across 67% and 44% of studies respectively. Sequence generation and allocation concealment had a low risk of bias assessment in about half of the studies. 50% of studies had unclear risk of bias for selective outcome reporting due to the lack of sufficient information.

[Insert Figure 2]

Trials of hand sanitiser or soap and water versus control

Figure 3a presents the meta-analysis of all trials, regardless of whether the number of handwashes could be estimated. Combining the five trials of soap and water hand hygiene versus control found a non-significant increase in ARI events: risk ratio 1.03 (95% CI 0.86 to 1.23) but with high heterogeneity. The nine trials of hand sanitiser versus control found a significant reduction in ARI events: risk ratio 0.85 (95% CI 0.77 to 0.94), providing some indirect evidence in favour of hand sanitiser.

A similar pattern of results was found when only trials for which the number of handwashes could be estimated were considered (Figure 3b), combining the three trials of soap and water hand hygiene versus control found a non-significant increase in ARI events: risk ratio 1.23 (95% CI 0.78 to 1.93) but with high heterogeneity (Figure 3b). Combining the seven trials of hand sanitiser versus control found a significant reduction in ARI events: risk ratio 0.80 (95% CI 0.71 to 0.89).

[Insert Figure 3a and 3b]

Dose-response relationship: hand hygiene frequency versus risk of respiratory infection (ARI, ILI or influenza)

Eleven of the trials provided sufficient information to estimate the dose of hand hygiene, which we converted to number of hand hygiene events per day. Plotted against the relative risk of ARIs, there is little dose-response relationship evident for hand sanitiser (Figure 4). The difference in effectiveness between hand sanitiser and soap and water does not appear to be explained by a difference in frequency. The cluster randomised trial by Little and

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colleagues [22] primarily used soap and water but also offered participants free hand sanitiser; only 18% report collecting the sanitiser.

Head-to-head trials of hand hygiene with hand sanitiser vs with soap and water

Four trials directly compared hand sanitiser with soap and water: two in childcare centres, one at a primary school, and one in workplaces. In a cluster randomised trial of children and staff in Swedish childcare centres, those at centres who were randomised to use an alcohol-based oily disinfectant gel (70% ethanol) after regular hand washing had a reduction in absenteeism rate of 12% (95% CI 4% to 20%) compared to control centres which used only soap and water [13]. The 3-arm cluster randomised trial of 24 childcare centres in Spain - educational and hand hygiene measures (one with soap and water; another with hand sanitiser) and a control group found children in the sanitiser group had a 13% lower (95% CI 6% to 28%) risk of respiratory infection than children in the soap and water group [12].

In Kenya, a cluster randomised trial assigned two primary schools to receive a handwashing with soap and water intervention, two to receive a sanitiser intervention, and two were a control [14]. Compared to control group students, both intervention groups had a reduction in observed rhinorrhoea (RR 0.77, 95% CI 0.62-0.95 for both sanitiser vs control and soap vs control). No significant differences between the sanitiser and soap groups were observed for respiratory outcomes. The 3-arm trial in six companies in Finland randomised workplaces to equip workplace bathrooms with liquid hand soap (soap and control arms) or alcohol-based hand rub [15]. Participants in the intervention arms also received guidance on additional strategies for limiting infection transmission. Before the onset of the 2009 influenza pandemic (and the subsequent national hand hygiene campaign), a statistically significant ($p = 0.002$) difference in the infection episodes was observed between the control (6.0 per year) and the soap-and-water arm (5.0 per year), but not between the control and the alcohol-rub arm (5.6 per year). Neither intervention had an effect on work absenteeism.

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Discussion

Based on both indirect and direct (head-to-head) trials, hand hygiene using alcohol-based hand sanitiser appears more effective at reducing ARI transmission than hand hygiene using with soap and water, with the difference in effect not explained by the difference in frequency of hand hygiene. This is an important finding, as most guidelines consider the two hand hygiene processes as equivalent, based on microbiological data for correctly performed processes.

The apparent greater effectiveness of hand sanitiser may be explained by its greater convenience, the lesser time required to perform hand hygiene, more sustained compliance with hand hygiene, and less irritation to the skin [30]. From our review of current research, we cannot determine the relative contribution of these behavioural elements, and hence further research is warranted to examine those contributions, and ways to improve each.

Limitations of this review are that conclusions are mostly from indirectness evidence, with direct evidence available from only four head-to-head trials, and that it was not possible to estimate the dose of hand hygiene for some trials. The variable duration of the interventions (which ranged from 2 to 26 months) in the included studies may have impacted the reported intervention adherence and hence the comparability of it.

A recent Cochrane review of the effect of rinse-free handwashing, compared to traditional hand hygiene, on absenteeism for ARI in preschool and school children reported a significant reduction in absenteeism of 9 days per 1000 available days for children in the rinse-free group, with the results coming from six randomised trials [9]. The effectiveness of handwashing with materials other than sanitiser or soap and water, such as ash, which may be used in low-income countries has mostly been examined in observational studies with uncertain effects [31].

Although the current evidence modestly favours hand sanitiser, further trials of hand hygiene methods are warranted. Any such trials should include a set of process measures looking at the “intervention fidelity” elements, such as frequency and correctness of hand hygiene processes. Meanwhile, policy documents and public guidance should continue to

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suggest both, but indicate that current evidence somewhat favours sanitiser for behavioural rather than biological reasons, and that it should be recommended where feasible.

Conclusions

Hand hygiene has a modest but important role in reducing the transmission of ARIs. Adequately performed hand hygiene, with either soap or sanitiser, reduces the risk of acute respiratory virus transmission. However, from both the direct and indirect comparisons in this review, sanitiser appears more effective in practice. While further head-to-head randomised trials are warranted, the current evidence appears sufficient to promote the use of hand sanitiser as the primary means for many everyday situations.

Abbreviations: Acute respiratory infections (ARI), Influenza-like illness (ILI), Risk Ratio (RR), Confidence interval (CI)

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Statements

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Contributors:

TH, PG, and CDM conceived the study. MB and NK screened studies for inclusion in this review and extracted the data. TH, AMS, PG and CDM analysed the data. TH wrote the first draft. All authors approved and revised the final manuscript.

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Competing interests:

None declared.

Ethical approval:

Ethical approval was not required.

Data sharing

All data relevant to the study are included in the article or uploaded as supplementary information.

Transparency

The corresponding authors affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

Patient and public involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

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Figure Caption List

Figure 1. PRISMA Flow Chart

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Figure 3a. Meta-analysis of all studies (regardless of whether the number of handwashes could be estimated) regardless of the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)

Figure 3b. Meta-analysis of studies whose number of handwashes could be estimated, subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)

Figure 4. Hand hygiene frequency ('dose') versus risk of respiratory infection (ARI, ILI or influenza)

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Supplementary data

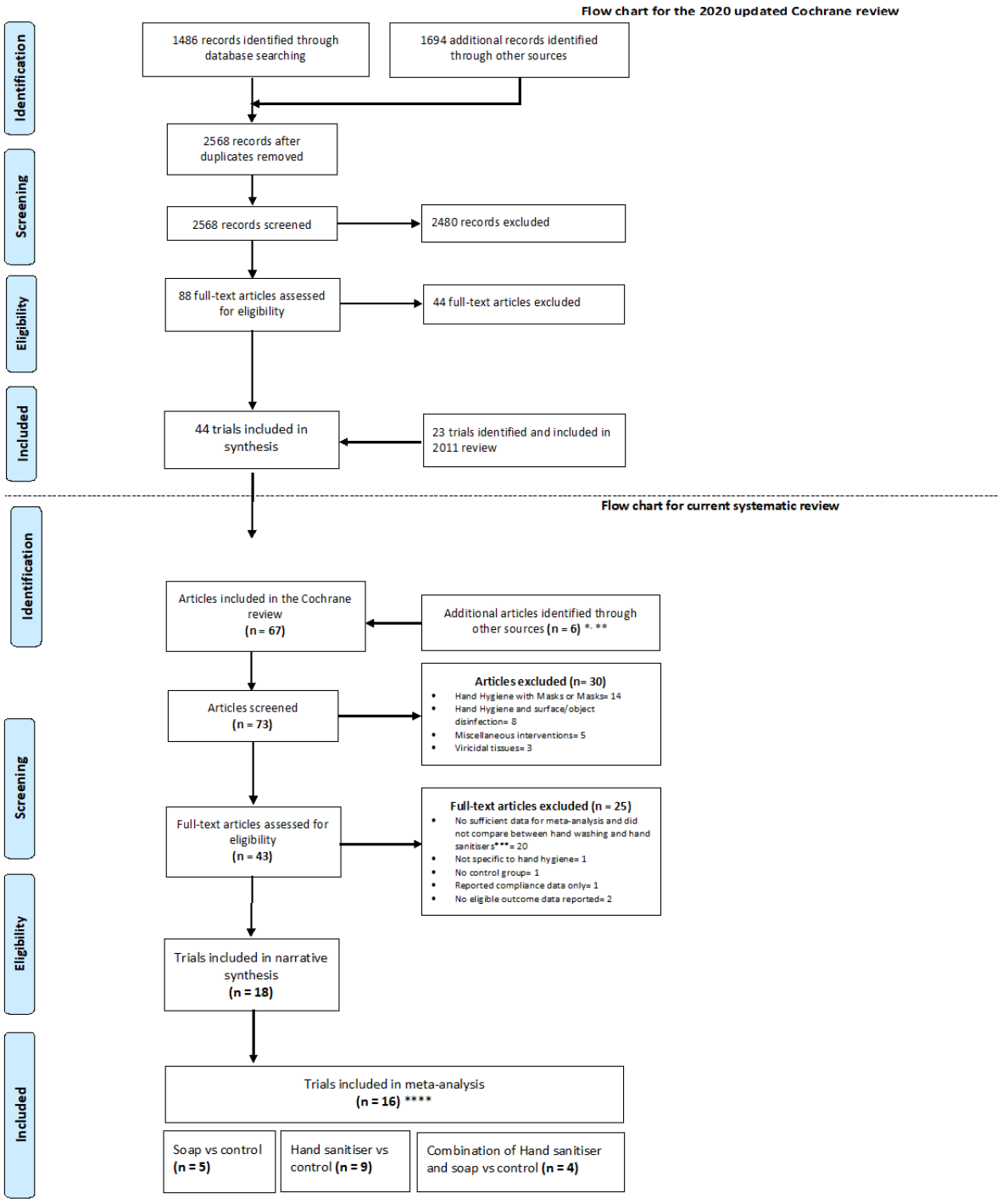
Online file 1 – Example of search strategy

Online file 2 – Summary of interventions and controls used in included studies

Online file 3 – Risk of bias assessment for each included trial

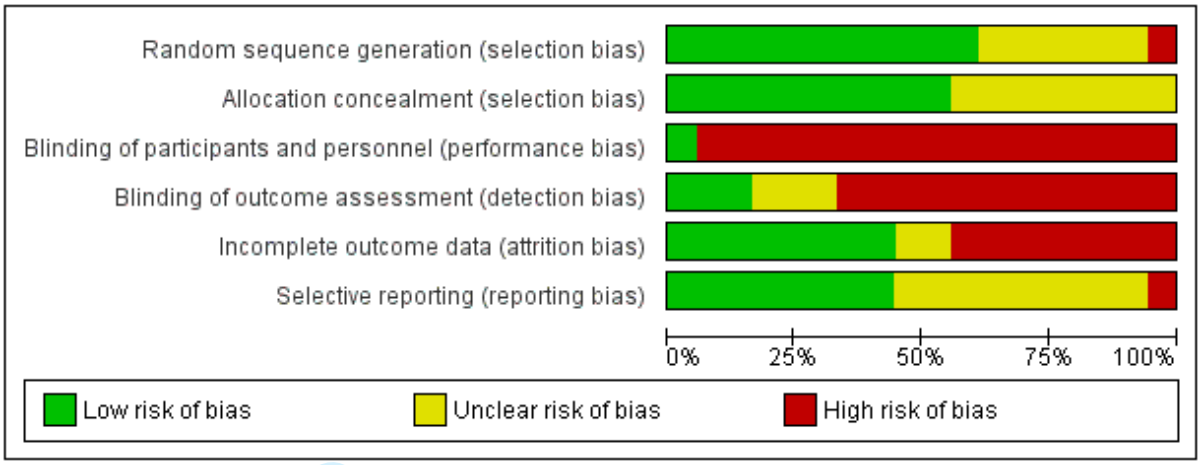
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Figure 1. PRISMA Flow Chart



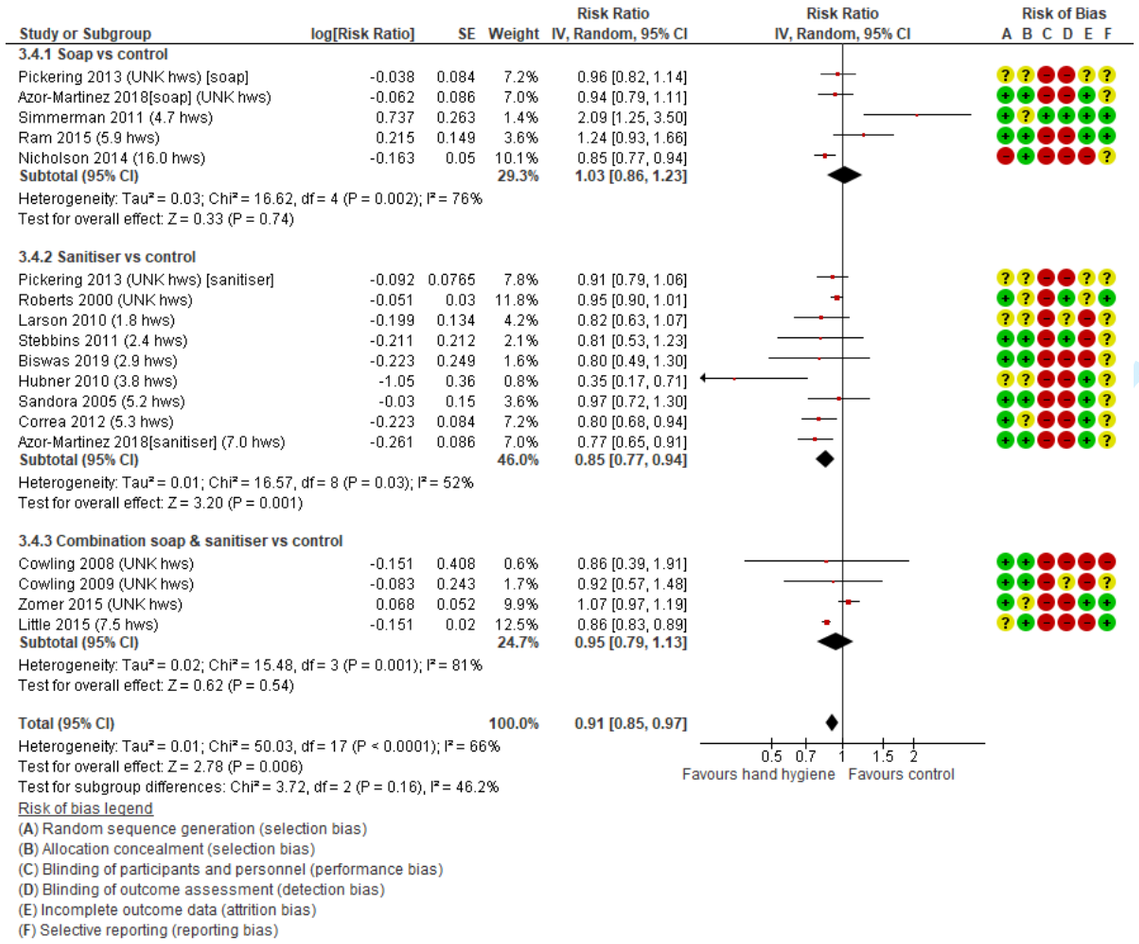
* Articles identified by screening the references of a newly published systematic review (Munn et al. 2020).
** One of the identified additional studies (Lennell et al. 2008) was excluded from the Cochrane review, because no respiratory illness data could be extracted. However, it was included in this review as it reported head to head comparison between hand washing and hand sanitiser and it contained absenteeism data which was an eligible outcome for this review.
*** Two studies did not have sufficient data for meta-analysis, but were included in the head to head comparison between hand washing and hand sanitiser (Lennell et al. 2008 & Savolainen-Kopra et al. 2012)
**** Two studies (Azor-Martinez et al. 2018 & Pickering et al. 2013) had 2 intervention groups. 1 group used soap and water, and another used hand sanitiser. Therefore, the number of studies, split by type of hand hygiene intervention, adds up to 18 and not 16 studies.

Figure 2. Overall risk of bias - presented as percentages across all included RCTs.



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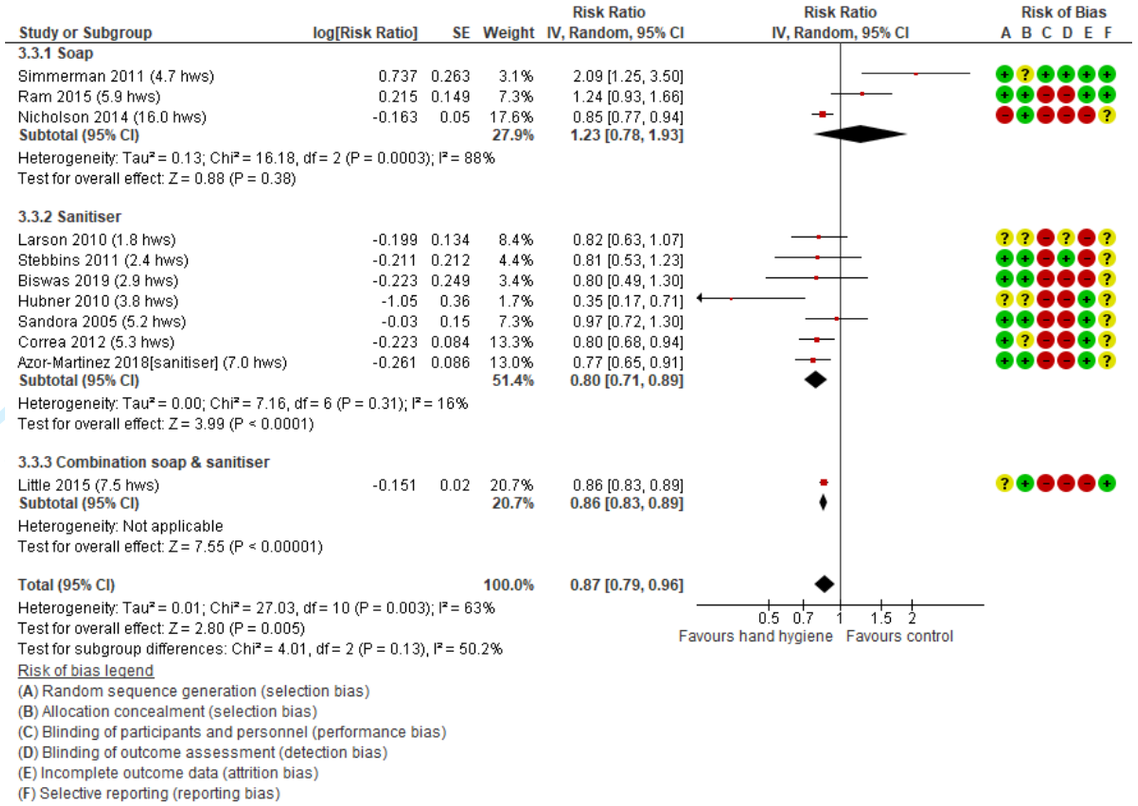
Figure 3a. Forest plot of meta-analysis of all trials (regardless of whether the number of handwashes could be estimated), subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)



Number of handwashes (hws) indicated in brackets after study reference where estimable. UNK = the number of handwashes was not estimable.

Figure 3b. Meta-analysis of studies whose number of handwashes could be estimated, subgrouped by the type of handwash material (soap vs sanitiser vs combination of sanitiser and soap)

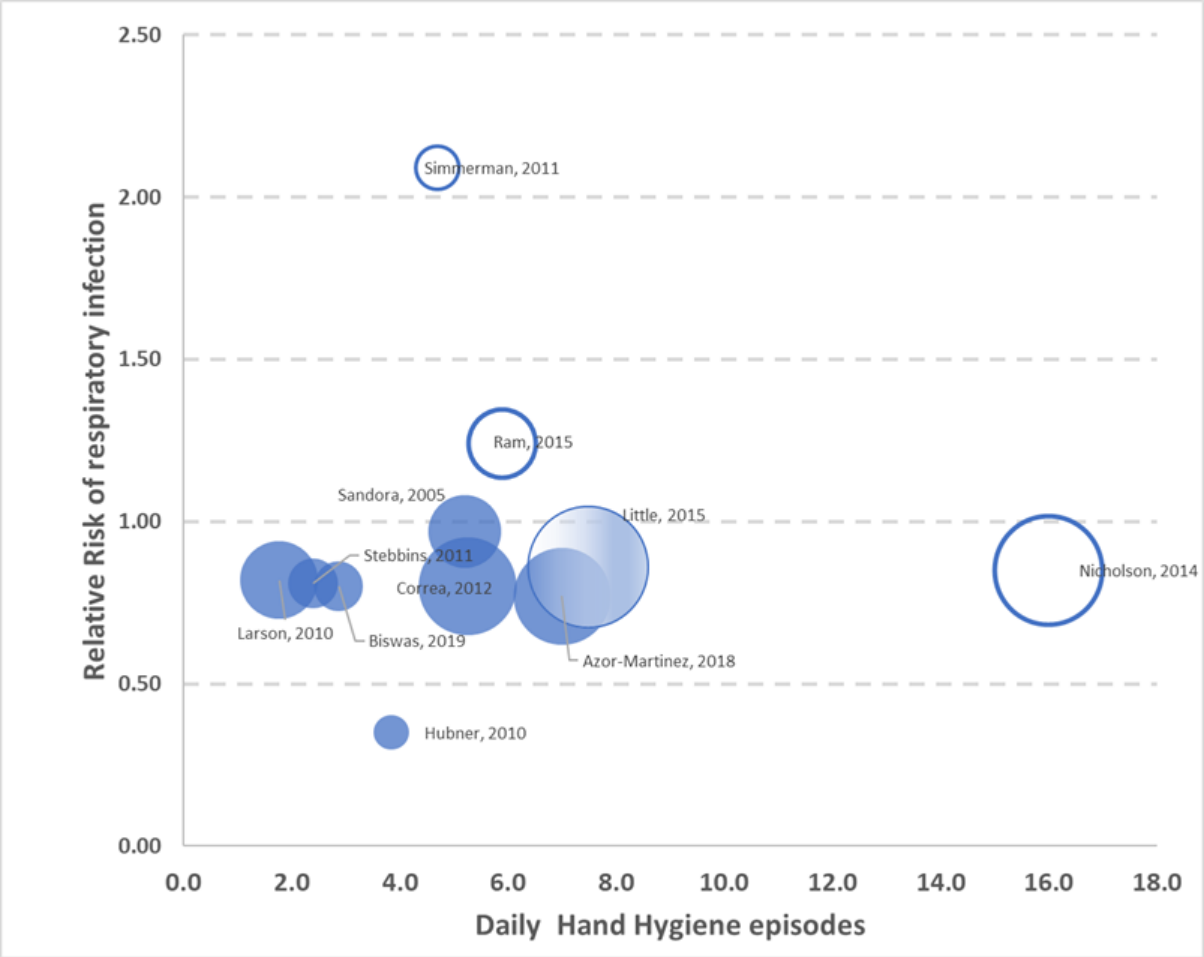
[Coloured figure in print]



Number of handwashes (hws) indicated in brackets after study reference where estimable.

Figure 4. Hand hygiene frequency ('dose') versus risk of respiratory infection (ARI, ILI or influenza)

[Coloured figure in print]



Clear circles = studies using soap

Coloured-in circles = studies using hand sanitiser

only

Online supplementary file 1: Search strings for databases

PubMed search string

("Influenza, Human"[Mesh] OR "Influenzavirus A"[Mesh] OR "Influenzavirus B"[Mesh] OR "Influenzavirus C"[Mesh] OR Influenza[tiab] OR "Respiratory Tract Diseases"[Mesh] OR "Bacterial Infections/transmission"[Mesh] OR Influenzas[tiab] OR “Influenza-like”[tiab] OR ILI[tiab] OR Flu[tiab] OR Flus[tiab] OR "Common Cold"[Mesh:NoExp] OR "common cold"[tiab] OR colds[tiab] OR coryza[tiab] OR coronavirus[Mesh] OR "sars virus"[Mesh] OR coronavirus[tiab] OR Coronaviruses[tiab] OR "coronavirus infections"[Mesh] OR "severe acute respiratory syndrome"[Mesh] OR "severe acute respiratory syndrome"[tiab] OR "severe acute respiratory syndromes"[tiab] OR sars[tiab] OR "respiratory syncytial viruses"[Mesh] OR "respiratory syncytial virus, human"[Mesh] OR "Respiratory Syncytial Virus Infections"[Mesh] OR "respiratory syncytial virus"[tiab] OR "respiratory syncytial viruses"[tiab] OR rsv[tiab] OR parainfluenza[tiab] OR “Respiratory illness”[tiab] OR ((Transmission[tiab]) AND (Coughing[tiab] OR Sneezing[tiab])) OR ((respiratory[tiab] AND Tract[tiab]) AND (infection[tiab] OR Infections[tiab] OR illness[tiab])))
AND
("Hand Hygiene"[Mesh] OR handwashing[tiab] OR hand-washing[tiab] OR ((Hand[tiab] OR Alcohol[tiab]) AND (wash[tiab] OR Washing[tiab] OR Cleansing[tiab] OR Rinses[tiab] OR hygiene[tiab] OR rub[tiab] OR Rubbing[tiab] OR sanitizer[tiab] OR sanitiser[tiab] OR cleanser[tiab] OR disinfected[tiab] OR Disinfectant[tiab] OR Disinfect[tiab] OR antiseptic[tiab] OR virucid[tiab])) OR "gloves, protective"[Mesh] OR Glove[tiab] OR Gloves[tiab] OR Masks[Mesh] OR "respiratory protective devices"[Mesh] OR facemask[tiab] OR Facemasks[tiab] OR mask[tiab] OR Masks[tiab] OR respirator[tiab] OR respirators[tiab] OR "Protective Clothing"[Mesh:NoExp] OR "Protective Devices"[Mesh] OR "patient isolation"[tiab] OR ((school[tiab] OR Schools[tiab]) AND (Closure[tiab] OR Closures[tiab] OR Closed[tiab])) OR Quarantine[Mesh] OR quarantine[tiab] OR “Hygiene intervention”[tiab] OR "Mouthwashes"[Mesh] OR gargling[tiab] OR “nasal tissues”[tiab] OR "Eye Protective Devices"[Mesh] OR Glasses[tiab] OR Goggle[tiab] OR “Eye protection”[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR “Face shield”[tiab] OR “Face shields”[tiab] OR Visors[tiab])
AND
("Communicable Disease Control"[Mesh] OR "Disease Outbreaks"[Mesh] OR "Disease Transmission, Infectious"[Mesh] OR "Infection Control"[Mesh] OR Transmission[sh] OR “Prevention and control”[sh] OR "Communicable Disease Control"[tiab] OR “Secondary transmission”[tiab] OR ((Reduced[tiab] OR Reduce[tiab] OR Reduction[tiab] OR Reducing[tiab] OR Lower[tiab]) AND (Incidence[tiab] OR Occurrence[tiab] OR Transmission[tiab] OR Secondary[tiab])))
AND

(Randomized controlled trial[pt] OR controlled clinical trial[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR "drug therapy"[sh] OR randomly[tiab] OR trial[tiab] OR groups[tiab])

NOT
(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))
NOT

(“Case Reports”[pt] OR Editorial[pt] OR Letter[pt] OR Meta-Analysis[pt] OR “Observational Study”[pt] OR “Systematic Review”[pt] OR “Case Report”[ti] OR “Case series”[ti] OR Meta-Analysis[ti] OR “Meta Analysis”[ti] OR “Systematic Review”[ti])

Cochrane Central Register of Controlled Trials (CENTRAL) search string

([mh "Influenza, Human"] OR [mh "Influenzavirus A"] OR [mh "Influenzavirus B"] OR [mh "Influenzavirus C"] OR Influenza:ti,ab OR [mh "Respiratory Tract Diseases"] OR Influenzas:ti,ab OR “Influenza-like”:ti,ab OR ILI:ti,ab OR Flu:ti,ab OR Flus:ti,ab OR [mh ^"Common Cold"] OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR [mh coronavirus] OR [mh "sars virus"] OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR [mh "coronavirus infections"] OR [mh "severe acute respiratory syndrome"] OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR [mh "respiratory syncytial viruses"] OR [mh "respiratory syncytial virus, human"] OR [mh "Respiratory Syncytial Virus Infections"] OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR “Respiratory illness”:ti,ab OR ((Transmission) AND (Coughing OR Sneezing)) OR ((respiratory:ti,ab AND Tract) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)))
AND
([mh "Hand Hygiene"] OR handwashing:ti,ab OR “hand-washing”:ti,ab OR ((Hand:ti,ab OR Alcohol:ti,ab) AND (wash:ti,ab OR Washing:ti,ab OR Cleansing:ti,ab OR Rinses:ti,ab OR hygiene:ti,ab OR rub:ti,ab OR Rubbing:ti,ab OR sanitizer:ti,ab OR sanitiser:ti,ab OR cleanser:ti,ab OR disinfected:ti,ab OR Disinfectant:ti,ab OR Disinfect:ti,ab OR antiseptic:ti,ab OR virucid:ti,ab)) OR [mh "gloves, protective"] OR Glove:ti,ab OR Gloves:ti,ab OR [mh Masks] OR [mh "respiratory protective devices"] OR facemask:ti,ab OR Facemasks:ti,ab OR mask:ti,ab OR Masks:ti,ab OR respirator:ti,ab OR respirators:ti,ab OR [mh ^"Protective Clothing"] OR [mh "Protective Devices"] OR "patient isolation":ti,ab OR ((school:ti,ab OR Schools:ti,ab) AND (Closure:ti,ab OR Closures:ti,ab OR Closed:ti,ab)) OR [mh Quarantine] OR quarantine:ti,ab OR "Hygiene intervention":ti,ab OR [mh Mouthwashes] OR gargling:ti,ab OR "nasal tissues":ti,ab OR [mh "Eye Protective Devices"] OR Glasses:ti,ab OR Goggle:ti,ab OR "Eye protection":ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab)
AND
([mh "Communicable Disease Control"] OR [mh "Disease Outbreaks"] OR [mh "Disease Transmission, Infectious"] OR [mh "Infection Control"] OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab))

Adapted from: T. Jefferson *et al.*, "Physical interventions to interrupt or reduce the spread of respiratory viruses," *Cochrane Database of Systematic Reviews,* 2020.

Online supplementary table 2: Summary of intervention and control for each included trial

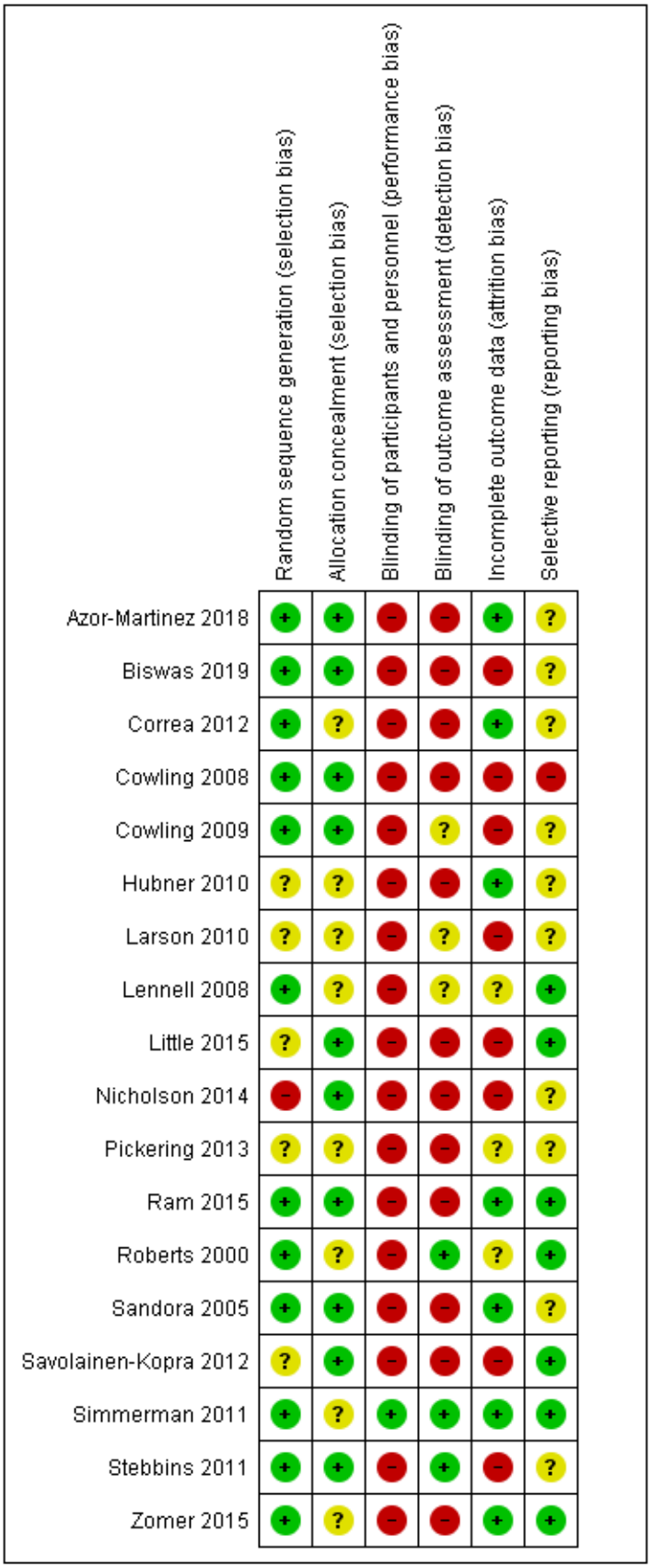
Study	Intervention	Control
Azor-Martinez 2018	1. Soap and water + hand hygiene workshop 2. Hand sanitiser + hand hygiene workshop	Existing/usual handwashing practices and education on hand hygiene
Biswas 2019	Hand sanitiser + respiratory hygiene education	Existing/usual handwashing practices
Correa 2012	Hand sanitiser as a complement to handwashing	Existing/usual handwashing practices
Cowling 2008	1. Healthy diet and lifestyle education + face mask education + face masks 2. Healthy diet and lifestyle education + hand hygiene education plus + sanitiser and soap	Education about the importance of a healthy diet and lifestyle for illness prevention and symptom alleviation
Cowling 2009	1. Healthy diet and lifestyle education + face mask education + face masks 2. Healthy diet and lifestyle education + hand hygiene education + hand sanitiser and soap	Education about the importance of a healthy diet and lifestyle for illness prevention and symptom alleviation
Hubner 2010	Hand sanitiser	Existing/usual handwashing practices
Larson 2010	1. Education on upper respiratory tract infection (URTI) prevention and treatment + hand-sanitiser 2. Education on URTI prevention and treatment + hand-sanitiser + face masks	Education on the prevention and treatment of URTIs and influenza
Lennell 2008	Hand sanitiser as a complement to handwashing	Existing/usual handwashing practices
Little 2015	Automated web-based educational intervention, monitored handwashing behaviour, automated emails and tailored feedback	Existing/usual handwashing practices - no access to intervention webpages
Nicholson 2014	Soap + social marketing programme	Existing/usual handwashing practices
Pickering 2013	1. Teacher hand hygiene training session + soap 2. Teacher hand hygiene training session + hand sanitiser	Existing/usual handwashing practices
Ram 2015	Soap + daily handwashing promotion	Existing/usual handwashing practices

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Roberts 2000	Handwashing programme and training for staff and children	Existing/usual handwashing practices
Sandora 2005	Hand sanitiser + hand-hygiene educational materials	Education on a healthy diet and good nutrition
Savolainen-Kopra 2012	1. Hand-hygiene with soap and water + standardised instructions on limiting infection transmission 2. Hand-hygiene with hand sanitiser	Existing/usual handwashing practices
Simmerman 2011	1. Hand washing 2. Hand washing + face masks	Education on nutrition, physical activity, and smoking cessation
Stebbins 2011	Training in hand and respiratory hygiene + hand sanitiser	Existing/usual handwashing practices
Zomer 2015	Hand hygiene products (paper towel dispensers, soap, hand sanitiser, hand cream) + training sessions and materials + reminder posters/stickers	Existing/usual handwashing practices

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Online supplementary file 3: Risk of bias for each item for each included trial



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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4, 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4, 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4, 5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5, 6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5, 6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4, 5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	-
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Fig 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Figures
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7, 8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Fig 2 & Appendix A1
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	-
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	10

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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